

Engineering Project Detailed Research Plan

Please complete the information/questions begun/seen below in red ink. Save your plan to your computer and upload a copy to your application Forms Folder.

Date: 01/08/26

Student Name: Amaar Chughtai

Project Title: Personalized Deep Learning Model for Closed-Loop 40 Hz Entrainment to Optimize Theta-Gamma Coupling in Alzheimer's

Parts of the generic engineering project are listed below with descriptions to the students in the boxes. Students may provide a detailed research plan by describing their specific project in response to each box below.

Engineering Goal: PROBLEM BEING ADDRESSED: All engineering projects solve a problem or fill a need. This goal should be a simple statement that describes the product being designed, the customer it is for and the problem or need it satisfies. Example "The goal is to design a solar powered lawn mower for inexpensive automated lawn care for homeowners"

My Project Goal is:

Existing 40 Hz auditory entrainment mechanisms are typically delivered on a fixed schedule, where patients are given continuous auditory stimulation, and then predetermined breaks. The issue is Alzheimer Disease patients' responses to the stimuli weaken over time and differ across people. The goal of this project is to design and test a closed-loop software controller that uses live EEG feedback of an Alzheimer's patient listening to a 40 Hz auditory stimulus to decide whether the stimulus should stay on or pause for rest, such that it maximizes theta-gamma phase amplitude coupling for the patient.

Design Criteria: Design criteria define the product's required performance . Examples: " It will have a minimum speed of 10 KPH", The output will be within 15% of the mean of the experimental data". "It must withstand 15 repetitions of a 10N impact" The International System of units (SI) required.

My Project Design Criteria are the following:

The controller must output a stimulation decision (on or rest) every 1.0 second. The entire computation process including feature extraction and model inference for the predicted phase amplitude coupling score and the policy decision should complete within 0.5 seconds on a normal CPU. The EEG data should be processed in a continuous sliding window of 2.0 seconds. The model will predict the theta-gamma band phase-amplitude coupling (PAC) score computed using theta phase in 4-8 Hz and gamma amplitude in a fixed band around 40 Hz. Additionally, the decision of the controller should last for at least 5.0 seconds before changing. Most importantly, via simulation, the closed-loop schedule must outperform the fixed schedule by increasing mean PAC during stimulation by $\geq 15\%$.

Constraints: Constraints are factors that limit the engineer's flexibility such as size, cost, and time limitations. Examples: "It must fit in a box no larger than 10x20x50 cm" "The maximum cost is \$50" "The software must run in real time on a Raspberry Pi"

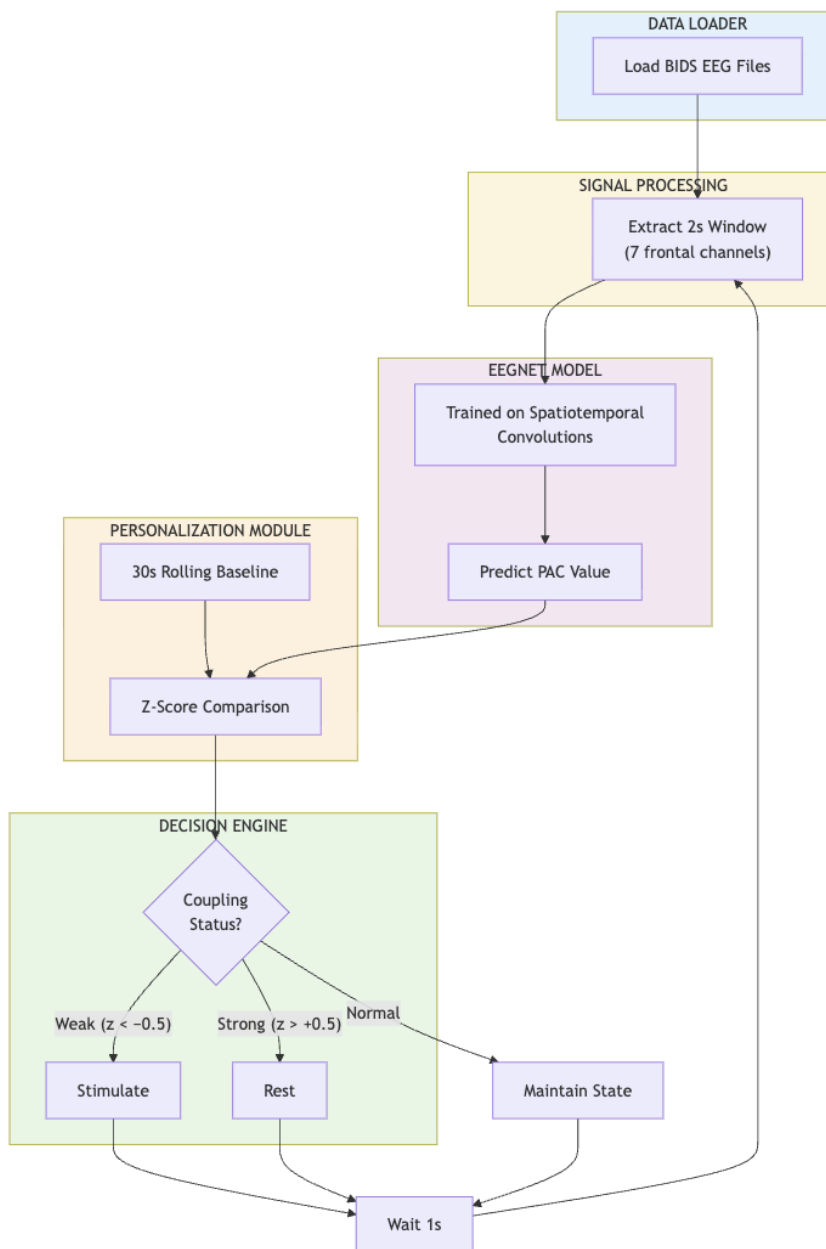
My Project Constraints are the following:

The project is limited to the public dataset OpenNeuro ds005048. The controller system is constrained to selecting only stimulation on or rest, and it cannot modify other stimulation parameters like frequency, amplitude, or waveform, the 40 Hz is fixed to the dataset protocol. The closed-loop model must carry out its prediction using only EEG samples up to the current time. Additionally, the closed-loop model must be trainable on my RTX 3080 GPU in reasonable time so I can test the model. The system must be able to run and compute in real time on a standard CPU in accordance with the time constraints in the design criteria specified above.

Provide your chosen design. For hardware, provide a sketch. For software, provide a flowchart. Indicate the components you will develop, and the libraries you are using.

My Project Design is shown below: insert photos, diagrams, or illustrations below.

The closed-loop controller consists of 5 main components. I will design a data loader to intake the BIDS EEG files, a signal processing unit for computing phase amplitude coupling labels, an EEGNet deep learning model for predicting PAC from 2 second windows of EEG, a personalization module implementing patient baseline adaptation, and a decision engine for stimulation control. I will be using PyTorch as the deep learning framework to build and train EEGNet. For loading the EEG, I will use MNE-Python. For the bandpass filtering for theta and gamma powers I will use SciPy, and NumPy will be used for almost all numerical operations. Pandas will handle the event files and allow me to visually display PAC accuracies. Together, the EEGNet model will receive 2.0 second windows from the 7 frontal electrode channels, predict PAC files, compare the prediction to the patient's 30 second baseline, and then output its stimulation decision when the coupling is beneath the patient's personalized threshold.



Test and evaluate your prototypes against the design criteria listed above to show how well the product meets the need/goal. Provide a test plan describing how you will test the design criteria and constraints you listed above., How will you analyze the data? If the product requires human testing please fill out and append <https://science-fair.org/wp/wp-content/uploads/2015/10/Research-Plan-Human-Participants.docx>

I test and analyze my prototypes using the following methods:

I will test the closed-loop controller by replaying pre-recorded EEG data from the OpenNeuro dataset. At each simulated decision point, the controller accesses only the data up to the time it is at, and outputs whether the auditory stimulus should be played or paused. To simulate how the brain responds to these decisions, I will extract the PAC response dynamics from the recorded 40 second stimulus periods and 20 second rest periods already in the dataset; when the controller decides to rest, I apply the observed rest response rate. I will run the simulation on 7 test subjects not used during EEGNet training to prevent overfitting. Most importantly, I will compare my controller's PAC optimization and total stimulation time against the scheduled stimulus' performance metrics that was used in the original study.

Bibliography: List at least five (5) major references (e.g. science journal articles, books, internet sites & dates of review) from your literature review. If you plan to use vertebrate animals, one of these references must be an animal care reference.

Example: Author's Name, Year of publication, "Quoted Title of Magazine Article (magazines only)"; Underlined *Title of Book or Magazine, date, volume, and number of magazine issue. Page numbers read.*

If you use a web site: www.urlname.ext, name of topic from the home page, author, and date read.

1. Shakya R, Suffczynski P, Shrestha S, Dangol S, Poudyal P, Mansur DI. 40 Hz binaural beats entrainment enhances the mood and cognition of medical students. *Int J Neurosci*. 2026 Jan;136(1):24-36. doi: 10.1080/00207454.2024.2429495. Epub 2024 Nov 19. PMID: 39535938. DOA 12/18/25
2. Martorell AJ, Paulson AL, Suk HJ, Abdurrob F, Drummond GT, Guan W, Young JZ, Kim DN, Kritskiy O, Barker SJ, Mangena V, Prince SM, Brown EN, Chung K, Boyden ES, Singer AC, Tsai LH. Multi-sensory Gamma Stimulation Ameliorates Alzheimer's-Associated Pathology and Improves Cognition. *Cell*. 2019 Apr 4;177(2):256-271.e22. doi: 10.1016/j.cell.2019.02.014. Epub 2019 Mar 14. PMID: 30879788; PMCID: PMC6774262. DOA 12/20/25
3. Lahijanian, M., Aghajan, H. & Vahabi, Z. Auditory gamma-band entrainment enhances default mode network connectivity in dementia patients. *Sci Rep* **14**, 13153 (2024). <https://doi.org/10.1038/s41598-024-63727-z> DOA 12/20/25
4. Tort AB, Komorowski R, Eichenbaum H, Kopell N. Measuring phase-amplitude coupling between neuronal oscillations of different frequencies. *J Neurophysiol*. 2010 Aug;104(2):1195-210. doi: 10.1152/jn.00106.2010. Epub 2010 May 12. Erratum in: *J Neurophysiol*. 2010 Oct;104(4):2302. PMID: 20463205; PMCID: PMC2941206. DOA 12/21/25
5. Yang YH, Chang HW, Chien CF, Hour TC. Investigating the effects of 40 Hz sound stimulation on Alzheimer's disease pathways: Modulation of amyloid- β 42 secretion, tau phosphorylation, phagocytosis, and autophagy. *J Alzheimers Dis*. 2025 Jul;106(1):184-

196. doi: 10.1177/13872877251339774. Epub 2025 May 4. PMID: 40320777. DOA
01/04/26